VIVA QUESTIONS

1. What is Bootstrap?

Answer: Bootstrap is a popular open-source and free front-end framework for developing responsive and mobile-first websites. It includes HTML, CSS, and JavaScript components.

2. What are the key features of Bootstrap?

Answer: Key features include a responsive grid system, pre-designed components like buttons and forms, JavaScript plugins, and customization options.

3. Explain the Bootstrap Grid System.

Answer: The grid system in Bootstrap is a 12-column layout that allows to create responsive designs by dividing the page into rows and columns. It adjusts based on screen size using breakpoints (e.g., `col-sm`, `col-md`).

4. What is the difference between `.container` and `.container-fluid`?

Answer: `.container` creates a fixed-width layout that adjusts based on screen size, while `.container-fluid` creates a full-width layout that spans the entire width of the viewport.

NOTE: containers are not nestable (cannot put a container inside another container)

5. How do you make a button in Bootstrap?

Answer: Use the `.btn` class along with a contextual class like `.btn-primary`, `.btn-secondary`, etc., to create a button, e.g., `<button class="btn btn-primary">Button</button>`.

6. What are Bootstrap breakpoints?

Answer: Breakpoints are predefined screen sizes in Bootstrap where the layout changes to ensure responsiveness. They include `xs` (extra small), `sm` (small), `md` (medium), `lg` (large), and `xl` (extra large).

7. What is a Bootstrap Jumbotron?

Answer: A Jumbotron is a large, attention-grabbing section used for showcasing important content on a website. It's created using the `.jumbotron` class.

8. How do you create a responsive image in Bootstrap?

Answer: Use the `.img-fluid` class to make an image responsive, ensuring it scales with the parent element while maintaining its aspect ratio.

9. What is a Bootstrap Navbar?

Answer: A Navbar is a responsive navigation header that adapts to different screen sizes. It's created using the `.navbar` class and can include links, brand logos, and other content.

10. How does Bootstrap handle form styling?

Answer: Bootstrap provides pre-styled form controls like inputs, text areas, checkboxes, and radios. It uses classes like `.form-control` for inputs and `.form-group` for grouping related form controls.

11. Who developed Bootstrap and when?

Answer: Bootstrap was developed by Mark Otto and Jacob Thornton at Twitter. It was initially released as an open-source project in August 2011.

12. Why was Bootstrap created?

Answer: Bootstrap was created to standardize the use of design elements across internal tools at Twitter, providing consistency and reducing the amount of repetitive code.

13. Where was Bootstrap first introduced?

Answer: Bootstrap was first introduced as an open-source project on GitHub, allowing developers worldwide to use and contribute to it.

14. What was the original name of Bootstrap?

Answer: Bootstrap was originally named "Twitter Blueprint" before it was renamed and released as Bootstrap.

15. How did Bootstrap become popular?

Answer: Bootstrap became popular due to its ease of use, responsive design features, and comprehensive documentation, making it accessible for developers of all skill levels.

16. Difference between pagination and scrolling?

Answer: Pagination divides content into multiple pages with navigation links, offering faster load times and clear content sections. Scrolling allows continuous content viewing, providing a seamless experience but can lead to longer load times.

17. What's responsive web design?

Answer: Responsive web design is an approach that ensures a website adapts its layout and elements to fit different screen sizes and devices, providing an optimal user experience on desktops, tablets, and smartphones.

18. Applications of bootstraps.

Answer:

-> **Responsive Web Design**: Ensures websites adapt to various screen sizes.

->**Pre-styled Components**: Provides ready-made UI elements like buttons, forms, and navbars.

->**Grid System**: Helps create structured, responsive layouts easily.

->**Cross-browser Compatibility**: Ensures consistent appearance across different browsers.

->**Customizable Themes**: Allows easy customization of design elements to match brand identity.

->**Quick Prototyping**: Speeds up the development process with reusable code and components.

->**Mobile-first Approach**: Designs are optimized for mobile devices by default.

19. Why Bootstrap?

- Responsive Design: Ensures websites work well on all screen sizes.
- Pre-designed Components: Speeds up development with ready-made UI elements.
- Mobile-first Approach: Prioritizes mobile device compatibility.
- Cross-browser Compatibility (browser support): Ensures consistent appearance across different browsers.
- Customizable: Easily tailored to fit specific design needs.
- Extensive Documentation: Offers detailed guides and examples for quick learning.
- Community Support: Large user base and active community for troubleshooting and resources.
- Easy to get started: with just the knowledge of html and css any one can started bootstraps.

20. What is Git?(global information tracker)

Answer: Git is a distributed version control system that allows multiple developers to work on a project simultaneously by tracking changes in the source code and facilitating collaboration.

21. What is GitHub?

Answer: GitHub is a cloud-based platform that hosts Git repositories, providing tools for version control, collaboration, and project management.

22. What is the difference between Git and GitHub?

Answer: Git is a version control system for tracking code changes locally, while GitHub is a platform that hosts Git repositories online, enabling collaboration and sharing.

GIT	GITHUB
Git is a software	**GitHub is a service**
Git is a command line tool	**GitHub is a graphical user interface**
Git is installed locally on the system	**GitHub is hosted on the web**
Git is focused on version control & code sharing	**GitHub is focused on centralized source code hosting**
Git is a version control system to manage source code history.	**GitHub is a hosting service for git repositories**
It was 1st released in 2005	**It was 1st launched in 2008**

23. What is a repository in Git?

Answer: A repository, or repo, is a storage space where your project's files and their revision history are stored. It can be local (on your computer) or remote (on platforms like GitHub).

24. How do you initialize a Git repository?

Answer: Use the command git init in the project directory to create a new Git repository.

25. What is a commit in Git?

Answer: A commit is a snapshot of your repository at a specific point in time. It records changes made to the files and includes a message describing the changes.

26. How do you create a new branch in Git?

Answer: Use the command git branch <branch-name> to create a new branch. To switch to that branch, use git checkout <branch-name>.

27. What is the purpose of branching in Git?

Answer: Branching allows you to work on different features or fixes separately without affecting the main codebase. It helps in parallel development and experimentation.

28. How do you merge branches in Git?

Answer: Use the command git merge <branch-name> while on the branch you want to merge into (usually the main branch) to combine the changes from another branch.

29. What is a pull request in GitHub?

Answer: A pull request is a GitHub feature that allows you to request the merging of changes from one branch to another. It enables code review and discussion before merging.

30. How do you clone a GitHub repository?

Answer: Use the command git clone <repository-URL> to create a copy of a remote repository on your local machine.

31. What is the purpose of git pull and git push?

Answer: git pull fetches and merges changes from a remote repository to your local branch, while git push uploads your local commits to a remote repository.

32. What is a .gitignore file?

Answer: A .gitignore file specifies files and directories that Git should ignore and not track, such as temporary files, build output, and sensitive information.

33. What is a fork in GitHub?

Answer: A fork is a copy of someone else's repository that you can modify independently. It's often used to contribute to open-source projects.

34. What is a Git stash?

Answer: git stash temporarily saves your changes without committing them, allowing you to switch branches or perform other tasks without losing your work.

35. where to get bootstraps?

Answer: there are 2 ways to start bootstraps:

i) Download bootstrap from getbootstrap.com
ii) Include bootstrap from CDN link (content delivery network)

36. What is front-end development?

Answer: Front-end development involves creating the visual and interactive aspects of a website or web application using HTML, CSS, and JavaScript. It focuses on the user interface and user experience.

37. **What is the difference between HTML and HTML5?**

Answer: HTML is the standard markup language for creating web pages, while HTML5 is the latest version that introduces new features like semantic elements (e.g., <header>, <footer>), multimedia support (e.g., <audio>, <video>), and APIs for offline storage.

38. What is CSS and why is it used?

Answer: CSS (Cascading Style Sheets) is a stylesheet language used to control the appearance and layout of HTML elements. It is used to style web pages, including font, colour, spacing, and positioning.

39. What is JavaScript and how is it used in front-end development?

Answer: JavaScript is a scripting language used to add interactivity and dynamic behaviour to web pages, such as form validation, animations, and handling events like clicks or keypresses.

40. What are CSS preprocessors?

Answer: CSS preprocessors, like Sass and Less, are tools that extend CSS with features like variables, nested rules, and mixins, making it easier to write and maintain complex stylesheets.

41. What is responsive web design?

Answer: Responsive web design is an approach that ensures a website adapts its layout and content to fit different screen sizes and devices, providing an optimal user experience across desktops, tablets, and smartphones.

42. What is a CSS framework? Can you name a few?

Answer: A CSS framework is a library of pre-written CSS code that provides a structure and common styles to build responsive and consistent web designs. Examples include Bootstrap, Foundation, and Bulma.

43. What is the purpose of JavaScript frameworks and libraries?

Answer: JavaScript frameworks (e.g., Angular, React, Vue.js) and libraries (e.g., jQuery) provide pre-written code that simplifies and speeds up the development of complex, dynamic web applications by offering reusable components and structure.

44. What is the Document Object Model (DOM)?

Answer: The DOM is a programming interface for web documents. It represents the structure of an HTML or XML document as a tree of objects, allowing developers to manipulate elements, attributes, and content dynamically using JavaScript.

45. Git commands:-

git init

- **Definition:** Initializes a new Git repository in the current directory.
- **Syntax:** $git init

git clone

- **Definition:** Creates a copy of an existing repository from a remote source.
- **Syntax:** $git clone <repository-URL>

git add

- **Definition:** Stages changes (files or directories) to be committed.
- **Syntax: $**git add <file/directory> or git add . (to add all changes)

git commit

- **Definition:** Records staged changes in the repository with a descriptive message.
- **Syntax:** git commit -m "commit message"

git status

- **Definition:** Shows the status of changes in the working directory and staging area.
- **Syntax:** git status

git push

- **Definition:** Uploads local commits to a remote repository.
- **Syntax:** git push <remote> <branch>

git pull

- **Definition:** Fetches and merges changes from a remote repository into the current branch.
- **Syntax:** git pull <remote> <branch>

git branch

- **Definition:** Lists, creates, or deletes branches.

- **Syntax:** git branch (to list) or git branch <branch-name> (to create)

git checkout

- **Definition:** Switches to a different branch or restores files to a previous state.
- **Syntax:** git checkout <branch-name> or git checkout <commit-hash>

git merge

- **Definition:** Combines changes from one branch into the current branch.
- **Syntax:** git merge <branch-name>

git log

- **Definition:** Displays the commit history of the repository.
- **Syntax**: git log

git rm

- **Definition:** Removes files from the working directory and the staging area.
- **Syntax:** git rm <file>

git config

- **Definition:** Configures Git settings such as username, email, and other preferences.
- **Syntax:** git config --global user.name "name"
- **Syntax:** git config --global user.email "email@gmail.com"

46. What is npm?

Answer: npm (Node Package Manager) is a package manager for JavaScript, allowing developers to install, manage, and share libraries or modules. Open source developers use npm to share software.

47. Features of npm.

Answer:

Package Management: Easily install, update, and uninstall JavaScript packages.

Dependency Management: Automatically handles and installs dependencies required by projects.

Version Control: Ensures consistent package versions across different environments.

Script Automation: Run custom scripts for tasks like testing, building, and deploying.

Global and Local Packages: Supports both global and project-specific package installations.

Registry Access: Provides access to the vast npm registry with thousands of open-source packages.

Custom Configuration: Allows configuration of settings for proxy, registry, and package scopes.

48. Difference between JavaScript and Node js.

Answer:

JavaScript	*Node js*
Environment: Runs in web browsers.	**Environment:** Runs on the server-side**.**
Usage: Primarily used for front-end web development.	**Usage:** Used for back-end development and building server-side applications.
Execution: Executes in the browser's JavaScript engine (e.g., V8 in Chrome).	**Execution:** Executes outside the browser using the V8 engine, enabling server-side scripting
Modules: Limited to browser-based modules like DOM manipulation.	**Modules:** Supports Common JS modules, file system access, and network-related modules.
APIs: Interacts with the browser's API.	**APIs:** Provides APIs for server operations like file handling, databases, and networking**.**

49. What is jQuery?

Answer: jQuery is a fast, lightweight JavaScript library that simplifies HTML document traversal, event handling, animation, and AJAX interactions.

50. What is jQuery and why is it used?

Answer: jQuery is a JavaScript library designed to simplify HTML DOM tree traversal, event handling, and animation. It is used to write less code while achieving more functionality compared to vanilla JavaScript.

51. How do you hide or show an element using jQuery?

Answer: You can hide an element using $('#elementID').hide(); and show it using $('#elementID').show();.

52. What is the difference between .ready() and .load() in jQuery?

Answer: .ready() is used to run code as soon as the DOM is fully loaded, while .load() is used to run code after all the content (including images) is fully loaded.

53. Responsive web design vs mobile design

Answer: Responsive Web Design vs. Mobile Design:

Responsive Web Design: Creates a single website that adjusts its layout and content to fit any screen size, from desktops to smartphones. It uses flexible grids, images, and CSS media queries.

Mobile Design: Specifically designs a separate website or app optimized only for mobile devices, focusing on touch interactions, smaller screens, and mobile-specific features.

In summary, responsive design adapts to all devices, while mobile design targets mobile devices exclusively.

54. jQuery vs. JavaScript:

Answer:

JavaScript: A programming language used for web development to create dynamic and interactive content directly in the browser.

jQuery: A lightweight JavaScript library that simplifies common tasks like DOM manipulation, event handling, and animations, allowing you to write less code with easier syntax.

In short, jQuery is a tool built on JavaScript to make it easier and faster to work with web pages.

55. Labels vs. Badges:

Labels: Used to highlight key information or tag content, typically larger and placed on forms or next to text to indicate status or category (e.g., "New", "Primary").

Badges: Small, usually round elements used to display counts or notifications, often attached to icons or navigation items (e.g., message count in an inbox).

In short, labels provide context or categorization, while badges indicate numbers or alerts.

56. Collapse vs. Modal:

Collapse: A feature that toggles the visibility of content with a sliding effect, typically used for expandable sections like FAQs or accordions.

Modal: A popup window that overlays the main content, used for displaying important information or forms that require user interaction.

In short, collapse is for toggling content visibility, while a modal is for displaying overlay popups.

57. npm vs. Node.js:

npm (Node Package Manager): A tool used to install, manage, and share JavaScript packages or libraries.

Node.js: A runtime environment that allows JavaScript to be run on the server-side, enabling back-end development.

In short, **npm** manages JavaScript packages, while **Node.js** runs JavaScript on servers.

58. CSS vs. Bootstrap:

CSS (Cascading Style Sheets): A language used to style and design web pages from scratch, offering full customization.

Bootstrap: A front-end framework built on CSS, providing pre-designed components and responsive layouts to speed up web development.

In short, **CSS** is for custom styling, while **Bootstrap** offers ready-made design tools built with CSS.

59. Inline Form vs. Horizontal Form:

Inline Form: Displays form fields and labels in a single line, side by side, ideal for compact, quick input.

Horizontal Form: Aligns labels to the left of the input fields in a two-column layout, suitable for forms where clear label-input association is needed.

In short, **inline forms** are compact, while **horizontal forms** offer a structured label-input layout.